Peter Schwendeman

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Education

University of Michigan, College of Engineering — Ann Arbor, Michigan
Major: Computer Science, Minor: Math and Physics (Overall GPA: 4.0/4.0) — *Aug 2023 - Expected May 2026*

Relevant Coursework: Machine Learning, Linear Algebra, Abstract Algebra, General Relativity, Complexity Theory, Cryptography, Data Structures and Algorithms, Thermodynamics, Optics, Lagrangian Mechanics, Reinforcement Learning (W26), Analysis (W26)

Awards: William J. Branstrom Freshmen Prize, Dean's List, Ginns International Scholarship, J. Sun and J. Wu Scholarship

Shanghai Jiao Tong University — Shanghai, China
Summer Exchange Student — Physics and Computer Science — *May 2024 - August 2024*

Relevant Coursework: Quantum Computing, Quantum Mechanics, Probability and Statistics

Publications

1. Nielsan S.*, Cetin E.* **Schwendeman P.***, Sun Q. Xu J., Tang Y. Learning to Orchestrate Agents in Natural Language with the Conductor. In proceedings of the *International Conference on Learning Representations (2026).* arxiv.org/abs/2512.04388
2. Xu J.*, Sun Q.*, **Schwendeman P.**, Nielsen S. Cetin E., Tang Y. Trinity: An Evolved LLM Coordinator. In proceedings of the *International Conference on Learning Representations (2026).* arxiv.org/abs/2512.04695
3. Deng K.*, **Schwendeman P.S.***, Guan Y. Learnable Diffusion Framework for Mouse V1 Neural Decoding. *Advanced Science* (2026). doi.org/10.1002/advs.202520220
4. Deng K., **Schwendeman P.S.**, Guan Y. Predicting single neuron responses of the primary visual cortex with deep learning model. *Advanced Science*, 11 (2024). doi.org/10.1002/advs.202305626
5. Liang D., Walker J., **Schwendeman P.S.**, Chandrashekar A., Ackermann R., Olsen K.F., Beck-Broichsitter M., Schwendeman S.P.. Effect of PLGA raw materials on in vitro and in vivo performance of drug-loaded microspheres. *Drug Delivery and Translational Research* (2024). doi.org/10.1007/s13346-024-01577-y
6. Glaser J., **Schwendeman P.S.**, Anderson J.A., Glotzer S.C. Unified memory in HOOMD-blue improves node-level strong scaling, *Computational Materials Science*, 173 (2020). doi.org/10.1016/j.commatsci.2019.109359

* indicates equal contribution

Experience

Michigan Medicine — Computational Medicine and Bioinformatics — Ann Arbor, Michigan
Undergraduate Researcher — Advised by Dr. Yuanfang Guan — *May 2023 - Current*

- Analyzed bio-plausibility of CNN-based models to predict neuron responses of mice to ImageNet images which achieved 15-30% gains in cross-subject inference metrics. Published work in *Advanced Science*[4].
- Designed latent-diffusion model with transformer backbone to reconstruct images from neuronal responses. Generated 80,000 synthetic image-response pairs and trained on a 95%-5% synthetic-real split, achieving state-of-the-art correlation and generation quality. Manuscript in submission at *Advanced Science*[3].

Sakana AI — Tokyo, Japan
Research Scientist Intern — Advised by Dr. Yujin Tang — *May 2025 - September 2025*

- Applied Group Relative Policy Optimization (GRPO) to train a 7B Conductor language model to coordinate language-based agents in solving complex reasoning tasks. Achieved state-of-the-art (SOTA) results on LiveCodeBench, GPQA-Diamond. Co-first authored ICLR paper detailing this work[1].
- Aided in designing Trinity, an evolutionary framework that discovers effective multi-agent roles, communication, and reasoning strategies automatically for math and coding tasks. Co-authored ICLR paper on this work[2].
- Wrote multi-agent scaffolding and routing baselines for both approaches, including MasRouter, Reflection, Smoothie, RouterDC, and Mixture-of-Agents.

Deque Systems, Inc. — Remote
Software Development Intern — Core API Team — *Jun 2022 - Sep 2023*

- Contributed to 6 web accessibility testing tools, including axe-core (1B+ downloads), using Python, JavaScript, TypeScript, and C# to help frontend developers build accessible websites.
- Extended functionality of Deque's accessibility linter from VS Code to Sublime Text (in Python) and JetBrains IDEs (in Scala), expanding the product's user base. Demoed projects to company leadership.

University of Michigan — Department of Chemical Engineering — Ann Arbor, Michigan
Research Assistant — Advised by Dr. Sharon Glotzer, Dr. Jens Glaser, Dr. Timothy Moore — *Oct 2017 - May 2023*

- Utilized CUDA unified memory between GPUs to improve performance of molecular dynamics simulation of crystallization, enabling faster large scale physical simulations. Published in *Computational Materials Science*[6].
- Updated C++ backend of Particle Simulation tool *HOOMD-blue* (1M+ downloads) to use *Heterogeneous-Compute Interface for Portability* (HIP) instead of CUDA, enabling multi-GPU tasks on both Nvidia and AMD GPUs.
- Investigated how patchy triangular nanoparticles self-assemble into host-guest structures, an experimental system for selective encapsulation. Developed Python algorithms to analyze these structures in simulation data (link).

Teaching Experience

EECS 445: Introduction to Machine Learning University of Michigan
Instructional Aide *January 2025 - Current*

- Developed homework, projects, and exams for 300+ students. Taught recitations of up to 60 students (recorded and watched by 200+). Provided support through office hours and online forums.

Projects

- **Poker AI** (link)
  Trained an AI agent to play a Poker variant at no cost using deep reinforcement learning (RL) and free Kaggle GPU hours. Achieved near SOTA performance for deep RL methods (exploitability of 50 milli-big blinds per game).
- **SugarSense** (link)
  Extracted important features from blood glucose data of sandbox diabetes patients with Dexcom API. Used State Space Model (SSM) to predict next two hours of blood sugars, achieving 0.95 test correlation for seen patients.

Skills

- **Programming Languages**: C, C++, C#, Python, Java, JavaScript, TypeScript, SQL, Scala, HTML, CSS, Bash, Perl
- **Frameworks**: PyTorch, TensorFlow, Keras, Hugging Face, TRL, Hydra, Numpy, Pandas, Scikit-learn, Matplotlib, CUDA, Django, NodeJS, React
- **Tools**: Unity, Godot, Docker, GIT, PostgreSQL, MySQL, SQLite
- **Machine Learning**: Transformers, Diffusion Models, CNNs, RNNs, VAEs, SSMs, Reinforcement Learning (Bandits, MDPs, POMDPs, MCTS, Q-learning, DQN, REINFORCE, DDPG, SAC, TRPO, PPO, DPO, GRPO)

Additional

- Violinist in the Michigan Pops Orchestra. Two time counselor for Interlochen orchestra camp. MSBOA honors soloist.
- Raised $2,500 from 27 donors for the Juvenile Diabetes Research Foundation. Edited diabetes educational materials.
- Limited conversational language proficiency in Chinese and Japanese.
- Interests: Go, badminton, traveling, skiing, violin, making video games (like Project Odyssey), composing music